



Embedded Software Development & Validation Engineer

QP Code: SSC/Q4603

Version: 2.0

NSQF Level: 5

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Qualification Pack

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SSC/Q4603: Embedded Software Development & Validation Engineer

Brief Job Description

An Embedded Software Development & Validation Engineer is responsible for the development of embedded software which includes determining the project requirements through coordination with the client and engineering teams, designing the software architecture, and running tests to ensure the functionality of embedded systems. The individual also carries out appropriate documentation concerning the development of embedded software.

Personal Attributes

Individual should be able to analyze the clients relevant business problems and resolve with relevant solution. They should be able to work independently and collaboratively. The individual should have domain-specific knowledge for effective embedded software development and carry out various responsibilities in the job role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N4601: Explain the Software Development Lifecycle \(SDLC\) and agile methodology](#)
2. [SSC/N4602: Elaborate on Hardware, Lab Familiarization and Advanced C](#)
3. [SSC/N4603: Explain Embedded OS Fundamentals, Driver, Applications, and Middleware](#)
4. [SSC/N4604: Describe the process of Verification & Validation](#)
5. [SSC/N4605: Explain Packaging, Release, and DevOps](#)
6. [SSC/N4606: Elaborate on embedded machine learning and Domain Applications](#)
7. [SSC/N9014: Maintain an inclusive, environmentally sustainable workplace](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development

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Country	India
NSQF Level	5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO- 2015/2512.0501
Minimum Educational Qualification & Experience	Pursuing 2nd year of UG ((Engineering/ BSc Electronics) and continuous education)) OR Pursuing 2nd year of 2-year diploma after 12th OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience in relevant field
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/03/2026
NSQC Approval Date	29/03/2023
Version	2.0
Reference code on NQR	QG-05-IT-01859-2024-V1.1-NASSCOM & V2
NQR Version	1.1

Remarks:

NA

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SSC/N4601: Explain the Software Development Lifecycle (SDLC) and agile methodology

Description

This unit covers different aspects of software development lifecycle and the application of agile methodology

Scope

The scope covers the following :

- SDLC Lifecycles and Frameworks
- Agile methodology
- Architecting and Designing the System
- Development and Operations
- Embedded Testing
- Application of Embedded Software

Elements and Performance Criteria

SDLC Lifecycles and Frameworks

To be competent, the user/individual on the job must be able to:

- PC1.** Explain what is SDLC and identify its phases
- PC2.** Describe different types of SDLC Models
- PC3.** Describe the estimation & planning process for software development

Agile methodology

To be competent, the user/individual on the job must be able to:

- PC4.** Explain the agile methodology
- PC5.** Explain the application of Scrum and Kanban in software development
- PC6.** Explain the differences among Test Driven Development (TDD), Behavior Driven Development (BDD), and Model Driven Development

Architecting and Designing the System

To be competent, the user/individual on the job must be able to:

- PC7.** Describe the process of architecting and designing the system
- PC8.** Explain the importance of reviews in software development
- PC9.** Explain the use of programming languages
- PC10.** Explain the importance and process of Static Code Analysis (SCA)
- PC11.** Explain the benefits of using a Code Repository, e.g. GitLab, Github, Bitbucket, SourceForge, to host an application's code source
- PC12.** Discuss the software build and deployment process
- PC13.** Elaborate on engineering measurement and metrics
- PC14.** Identify the different roles and responsibilities in a formal review

Development and Operations

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To be competent, the user/individual on the job must be able to:

PC15. Explain the DevOps process flow

Embedded Testing

To be competent, the user/individual on the job must be able to:

PC16. Explain the meaning and process of embedded testing

Application of Embedded Software

To be competent, the user/individual on the job must be able to:

PC17. Explain the functioning of a digital door locking system

PC18. Describe the process of developing a mobile phone

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. SDLC and its phases

KU2. different types of SDLC Models

KU3. the estimation & planning process for software development

KU4. the agile methodology

KU5. the application of Scrum and Kanban in software development

KU6. the differences among Test Driven Development (TDD), Behavior Driven Development (BDD), and Model Driven Development

KU7. the process of architecting and designing the system

KU8. the importance of reviews in software development

KU9. the use of programming languages

KU10. the importance and process of Static Code Analysis (SCA)

KU11. the benefits of using a Code Repository, e.g. GitLab, Github, Bitbucket, SourceForge, to host an application's code source

KU12. the software build and deployment process

KU13. the engineering measurement and metrics

KU14. the different roles and responsibilities in a formal review

KU15. the DevOps process flow

KU16. the meaning and process of embedded testing

KU17. the functioning of a digital door locking system

KU18. the process of developing a mobile phone

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work-related notes and records

GS2. listen attentively to understand the information/ instructions being shared by the speaker

GS3. communicate clearly and politely with co-workers and clients

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- GS4.** coordinate with co-workers to achieve work objectives
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify possible disruptions to work and take appropriate preventive measures
- GS7.** take quick decisions to deal with workplace emergencies/ accidents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>SDLC Lifecycles and Frameworks</i>	4	13	-	-
PC1. Explain what is SDLC and identify its phases	1	4	-	-
PC2. Describe different types of SDLC Models	2	5	-	-
PC3. Describe the estimation & planning process for software development	1	4	-	-
<i>Agile methodology</i>	4	12	-	-
PC4. Explain the agile methodology	1	4	-	-
PC5. Explain the application of Scrum and Kanban in software development	1	4	-	-
PC6. Explain the differences among Test Driven Development (TDD), Behavior Driven Development (BDD), and Model Driven Development	2	4	-	-
<i>Architecting and Designing the System</i>	11	33	-	-
PC7. Describe the process of architecting and designing the system	2	4	-	-
PC8. Explain the importance of reviews in software development	1	4	-	-
PC9. Explain the use of programming languages	1	4	-	-
PC10. Explain the importance and process of Static Code Analysis (SCA)	2	4	-	-
PC11. Explain the benefits of using a Code Repository, e.g. GitLab, Github, Bitbucket, SourceForge, to host an application's code source	2	5	-	-
PC12. Discuss the software build and deployment process	1	4	-	-
PC13. Elaborate on engineering measurement and metrics	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Identify the different roles and responsibilities in a formal review	1	4	-	-
<i>Development and Operations</i>	2	5	-	-
PC15. Explain the DevOps process flow	2	5	-	-
<i>Embedded Testing</i>	1	4	-	-
PC16. Explain the meaning and process of embedded testing	1	4	-	-
<i>Application of Embedded Software</i>	3	8	-	-
PC17. Explain the functioning of a digital door locking system	1	4	-	-
PC18. Describe the process of developing a mobile phone	2	4	-	-
NOS Total	25	75	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4601
NOS Name	Explain the Software Development Lifecycle (SDLC) and agile methodology
Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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SSC/N4602: Elaborate on Hardware, Lab Familiarization and Advanced C

Description

This unit is about the use of relevant hardware in a lab environment, C library functions and applicable coding practices.

Scope

The scope covers the following :

- Embedded Systems
- Electrostatic Discharge Essentials
- Booting for Embedded Processor
- Securing Embedded Systems
- Building an executable
- Debugging and Tracing
- Memory Management
- Advanced Data Types
- Referencing data and functions
- Working with registers
- Context management
- C library functions
- Coding practices

Elements and Performance Criteria

Embedded Systems

To be competent, the user/individual on the job must be able to:

- PC1.** Explain what are embedded systems
- PC2.** Explain the basic embedded system architecture
- PC3.** Elaborate on standard interfaces in embedded systems
- PC4.** Explain how to understand schematics/datasheet

Electrostatic Discharge Essentials

To be competent, the user/individual on the job must be able to:

- PC5.** Explain what ESD is and why we deal with it
- PC6.** Identify the causes of ESD
- PC7.** Explain how ESD can occur at all levels of integration
- PC8.** Explain how to prevent ESD

Booting for Embedded Processor

To be competent, the user/individual on the job must be able to:

- PC9.** Explain the host and target development setups
- PC10.** Explain the functioning of system boot components

Securing Embedded Systems

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To be competent, the user/individual on the job must be able to:

PC11. Explain what is hardware security in embedded systems

PC12. Identify and explain the appropriate techniques to improve embedded systems

Building an executable

To be competent, the user/individual on the job must be able to:

PC13. Explain the role of linkers and memory layouts in building an executable

PC14. Elaborate on map file basics and explain how to read map file, symbols and object files

PC15. Explain the process of building libraries

Debugging and Tracing

To be competent, the user/individual on the job must be able to:

PC16. Explain how to debug with gdb

PC17. Explain the importance of including traces in code, passing values, capturing core dump and analysis

PC18. Explain the process of disassembling C code

Memory Management

To be competent, the user/individual on the job must be able to:

PC19. Explain the memory architecture

PC20. Describe the process of memory allocation

PC21. Explain the importance and process of memory profiling

Advanced Data Types

To be competent, the user/individual on the job must be able to:

PC22. Explain the variable length data structures

PC23. Elaborate on structures and unions

Referencing data and functions

To be competent, the user/individual on the job must be able to:

PC24. Compare hard real time with soft real time

PC25. Explain the functioning of traversing lists with pointers

PC26. Explain the process of dynamic binding

Working with registers

To be competent, the user/individual on the job must be able to:

PC27. Describe the bit level operations

PC28. Explain the handling of special registers

Context management

To be competent, the user/individual on the job must be able to:

PC29. Describe the process of bringing up CPU

PC30. Explain the requirements of C startup

PC31. Explain what execution context is

PC32. Explain the interrupt context

C library functions

To be competent, the user/individual on the job must be able to:

PC33. Explain the functioning of useful primitives

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PC34. Explain the system calls

PC35. Explain the OOPS concept

PC36. Describe the process of interprocess communications

PC37. Explain the importance of referring to implementations

Coding practices

To be competent, the user/individual on the job must be able to:

PC38. Explain the coding guidelines

PC39. Explain the importance of secure and safe Coding

PC40. Describe the process of developing the optimal code

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. what embedded systems are

KU2. the basic embedded system architecture

KU3. standard interfaces in embedded systems

KU4. schematics/datasheet in embedded software development

KU5. ESD and why it is dealt with

KU6. the causes of ESD and how to prevent it

KU7. how ESD can occur at all levels of integration

KU8. the host and target development setups

KU9. the functioning of system boot components

KU10. hardware security in embedded systems

KU11. the appropriate techniques to improve embedded systems

KU12. the role of linkers and memory layouts in building an executable

KU13. map file basics and how to read map file, symbols and object files

KU14. the process of building libraries

KU15. how to deb with gdb

KU16. the importance of including traces in code, passing values, capturing core dump and analysis

KU17. the process of disassembling C code

KU18. the memory architecture

KU19. the process of memory allocation

KU20. the importance and process of memory profiling

KU21. the variable length data structures

KU22. the structures and unions

KU23. hard real time vs. soft real time

KU24. the functioning of traversing lists with pointers

KU25. the process of dynamic binding

KU26. the bit level operations

KU27. the handling of special registers

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- KU28.** the process of bringing up CPU
- KU29.** the requirements of C startup
- KU30.** the execution context
- KU31.** the interrupt context
- KU32.** the functioning of useful primitives
- KU33.** the system calls
- KU34.** the OOPS concept
- KU35.** the process of interprocess communications
- KU36.** the importance of referring to implementations
- KU37.** the coding guidelines
- KU38.** the importance of secure and safe Coding
- KU39.** the process of developing the optimal code

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** listen attentively to understand the information/ instructions being shared by the speaker
- GS3.** communicate clearly and politely with co-workers and clients
- GS4.** coordinate with co-workers to achieve work objectives
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify possible disruptions to work and take appropriate preventive measures
- GS7.** take quick decisions to deal with workplace emergencies/ accidents
- GS8.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Embedded Systems</i>	3	7	-	-
PC1. Explain what are embedded systems	1	2	-	-
PC2. Explain the basic embedded system architecture	-	2	-	-
PC3. Elaborate on standard interfaces in embedded systems	1	2	-	-
PC4. Explain how to understand schematics/datasheet	1	1	-	-
<i>Electrostatic Discharge Essentials</i>	3	7	-	-
PC5. Explain what ESD is and why we deal with it	1	2	-	-
PC6. Identify the causes of ESD	-	2	-	-
PC7. Explain how ESD can occur at all levels of integration	1	1	-	-
PC8. Explain how to prevent ESD	1	2	-	-
<i>Bootling for Embedded Processor</i>	2	4	-	-
PC9. Explain the host and target development setups	1	2	-	-
PC10. Explain the functioning of system boot components	1	2	-	-
<i>Securing Embedded Systems</i>	1	4	-	-
PC11. Explain what is hardware security in embedded systems	1	2	-	-
PC12. Identify and explain the appropriate techniques to improve embedded systems	-	2	-	-
<i>Building an executable</i>	2	5	-	-
PC13. Explain the role of linkers and memory layouts in building an executable	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Elaborate on map file basics and explain how to read map file, symbols and object files	1	2	-	-
PC15. Explain the process of building libraries	-	2	-	-
<i>Debugging and Tracing</i>	2	5	-	-
PC16. Explain how to debug with gdb	1	2	-	-
PC17. Explain the importance of including traces in code, passing values, capturing core dump and analysis	-	1	-	-
PC18. Explain the process of disassembling C code	1	2	-	-
<i>Memory Management</i>	1	6	-	-
PC19. Explain the memory architecture	-	2	-	-
PC20. Describe the process of memory allocation	1	2	-	-
PC21. Explain the importance and process of memory profiling	-	2	-	-
<i>Advanced Data Types</i>	1	4	-	-
PC22. Explain the variable length data structures	-	2	-	-
PC23. Elaborate on structures and unions	1	2	-	-
<i>Referencing data and functions</i>	3	5	-	-
PC24. Compare hard real time with soft real time	1	2	-	-
PC25. Explain the functioning of traversing lists with pointers	1	2	-	-
PC26. Explain the process of dynamic binding	1	1	-	-
<i>Working with registers</i>	1	4	-	-
PC27. Describe the bit level operations	-	2	-	-
PC28. Explain the handling of special registers	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Context management</i>	2	8	-	-
PC29. Describe the process of bringing up CPU	-	2	-	-
PC30. Explain the requirements of C startup	-	2	-	-
PC31. Explain what execution context is	1	2	-	-
PC32. Explain the interrupt context	1	2	-	-
<i>C library functions</i>	3	10	-	-
PC33. Explain the functioning of useful primitives	-	2	-	-
PC34. Explain the system calls	1	2	-	-
PC35. Explain the OOPS concept	-	2	-	-
PC36. Describe the process of interprocess communications	1	2	-	-
PC37. Explain the importance of referring to implementations	1	2	-	-
<i>Coding practices</i>	1	6	-	-
PC38. Explain the coding guidelines	-	2	-	-
PC39. Explain the importance of secure and safe Coding	-	2	-	-
PC40. Describe the process of developing the optimal code	1	2	-	-
NOS Total	25	75	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4602
NOS Name	Elaborate on Hardware, Lab Familiarization and Advanced C
Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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SSC/N4603: Explain Embedded OS Fundamentals, Driver, Applications, and Middleware

Description

This unit covers the basics of embedded OS, memory, task and resource management, advance RTOS concepts, driver development, testing, etc.

Scope

The scope covers the following :

- Basics of Embedded OS
- Concept of Real-time OS
- Memory Management
- Task Management
- Inter Task (Process) Communication (synchronization)
- Resource Management
- Timer Management
- Task Notifications
- Advance RTOS concepts
- RTOS File System (freeRTOS+FAT)
- RTOS Project
- Introduction to ARM Cortex-M and its Architecture
- Interrupts, Timers and counters
- Driver development for a baremetal environmen
- Linux driver development and testing
- Optimizations

Elements and Performance Criteria

Basics of Embedded OS

To be competent, the user/individual on the job must be able to:

- PC1.** Elaborate on freeRTOS
- PC2.** Explain the process of downloading and installing freeRTOS
- PC3.** Describe the use of Compilation and build tool setup
- PC4.** Explain the simulator/ emulator requirements
- PC5.** Elaborate on hardware setup and demo
- PC6.** Explain freeRTOS source code organization
- PC7.** Explain the use of freeRTOS ports
- PC8.** Describe the use of demo applications
- PC9.** Describe the process of configuring freeRTOS
- PC10.** Describe the process of building a simple demo application
- PC11.** Explain the applicable simulator/ emulator requirements

Concept of Real-time OS

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To be competent, the user/individual on the job must be able to:

- PC12.** Compare hard real time with soft real time
- PC13.** Elaborate on Multi-threading/ Multi-tasking / Concurrent execution
- PC14.** Explain the need for scheduler
- PC15.** Explain context switching
- PC16.** Identify the features of freeRTOS
- PC17.** Explain the value proposition of freeRTOS

Memory Management

To be competent, the user/individual on the job must be able to:

- PC18.** Elaborate on advance C covering memory management
- PC19.** Identify different types of memory available
- PC20.** Elaborate on freeRTOS Heap Memory Management (dynamic vs static, predictability, fragmentation, different memory allocation schemes)
- PC21.** Explain the freeRTOS Heap Utility Functions
- PC22.** Describe the process of optimizing memory
- PC23.** Describe the process of debugging memory
- PC24.** Elaborate on memory map
- PC25.** Explain the relevant concepts - architecture flat memory model, no distinction between kernel and user memory in RTOS, memory protection unit etc.

Task Management

To be competent, the user/individual on the job must be able to:

- PC26.** Explain the concept of freeRTOS Tasks
- PC27.** Elaborate on freeRTOS Tasks APIs
- PC28.** Describe the process of creating tasks and setting task priorities
- PC29.** Elaborate on task state transitions
- PC30.** Elaborate on idle task and scheduling algorithms
- PC31.** Elaborate on time measurement and tick interrupt

Inter Task (Process) Communication (synchronization)

To be competent, the user/individual on the job must be able to:

- PC32.** Elaborate on Queue and freeRTOS Queue APIs
- PC33.** Explain the use of data storage for Queue (data of fixed length, variable length, mem copy vs by reference etc.)
- PC34.** Elaborate on blocking read, and write in Queue
- PC35.** Describe the process of receiving data from multiple queues
- PC36.** Elaborate on Events and Interrupt Service Routines (ISRs)
- PC37.** Conduct Tasks vs ISRs comparison
- PC38.** Describe the process of Nested Interrupt handling
- PC39.** Explain the concept of semaphores
- PC40.** Explain the synchronization mechanisms of binary Semaphores
- PC41.** Explain the use of counting semaphores

Resource Management

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To be competent, the user/individual on the job must be able to:

- PC42.** Explain the problem of shared resource corruption (multi-tasking system)
- PC43.** Describe the read, modify, write operations
- PC44.** Elaborate on Atomic operations with respect to resource management
- PC45.** Explain re-entrant functions or thread safe functions
- PC46.** Explain the concept and conditions of Mutual Exclusion
- PC47.** Elaborate on Critical Section with respect to shared resources
- PC48.** Elaborate on Critical Section
- PC49.** Explain the Mutex (locking) mechanism
- PC50.** Elaborate on deadlocks and starvation in programming
- PC51.** Explain Priority inversion
- PC52.** Elaborate on Priority Inheritance
- PC53.** Explain the utility of gatekeeper task (for guarding shared resources)
- PC54.** Explain the benefit of event groups for multiple task synchronization
- PC55.** Elaborate on event groups for multiple task synchronization

Timer Management

To be competent, the user/individual on the job must be able to:

- PC56.** Elaborate on the context and benefit of Software Timers
- PC57.** Describe the process of creating software timer
- PC58.** Elaborate on the requirement and process of resetting of software timer
- PC59.** Explain the use of hardware timers and watchdogs

Task Notifications

To be competent, the user/individual on the job must be able to:

- PC60.** Conduct tasks communication objects vs direct task notifications comparison
- PC61.** Explain the benefits of task notifications

Advance RTOS concepts

To be competent, the user/individual on the job must be able to:

- PC62.** Explain the use of SMP port of freeRTOS
- PC63.** Explain the benefits of memory protection unit support
- PC64.** Explain the requirement of low power support (Ticket execution feature)
- PC65.** Elaborate on trace and runtime stats
- PC66.** Explain the requirement of refresh on device drivers and integration
- PC67.** Conduct comparison of different popular RTOSes

RTOS File System (freeRTOS+FAT)

To be competent, the user/individual on the job must be able to:

- PC68.** Explain the file system concept
- PC69.** Elaborate on FreeRTOS + FAT (FAT12, FAT16, FAT32 DOS/Windows compatible File System)
- PC70.** Elaborate on File System APIs, source code organization, and configuration
- PC71.** List different examples of FAT, e.g. basic file use, stdio API tests, creating a disk etc.
- PC72.** List examples of FTP server and HTTP Web Server

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RTOS Project

To be competent, the user/individual on the job must be able to:

- PC73.** Explain system design from the perspective of hardware, tasks, interrupts/events, timers, and real-time processing
- PC74.** Identify the common problems encountered (crash, ISR issues, stack overflow issues), debugging (asserts, traces/logs), and profiling, etc.
- PC75.** Describe project delivery based on freeRTOS and embedded hardware board or emulator

Introduction to ARM Cortex-M and its Architecture

To be competent, the user/individual on the job must be able to:

- PC76.** Explain how to configure and run simple programs on the emulator and on the board
- PC77.** Elaborate on target board and GPIO programming

Interrupts, Timers and counters

To be competent, the user/individual on the job must be able to:

- PC78.** Explain the need of Interrupts
- PC79.** Elaborate on the need for Generic Clock tree, CPU interrupt flow, and Interrupt flow
- PC80.** Elaborate on Interrupt Flow – inside ISR, and Hardware ISR's – Coding practice

Driver development for a baremetal environmen

To be competent, the user/individual on the job must be able to:

- PC81.** Describe the process of developing and testing a simple serial Driver

Linux driver development and testing

To be competent, the user/individual on the job must be able to:

- PC82.** Elaborate on the need of Kernel drivers
- PC83.** Explain the benefits of Loadable modules
- PC84.** Describe the process of writing a Hello World Device driver
- PC85.** Describe the configuration of a USB device in Windows/Linux environment
- PC86.** Explain what a network driver is
- PC87.** Elaborate on the configurability of Network drivers (e.g. how SSID is configured for Wi-Fi, or Ethernet is put in promiscuous mode)

Optimizations

To be competent, the user/individual on the job must be able to:

- PC88.** Explain what optimization is
- PC89.** Describe different optimization methods

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the process of downloading and installing freeRTOS
- KU2.** the use of Compilation and build tool setup
- KU3.** the applicable simulator/ emulator requirements
- KU4.** the hardware setup and demo
- KU5.** freeRTOS source code organization

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- KU6.** the use of freeRTOS ports
- KU7.** the use of demo applications
- KU8.** the process of configuring freeRTOS
- KU9.** the process of building a simple demo application
- KU10.** the applicable simulator/ emulator requirements
- KU11.** Multi-threading/ Multi-tasking / Concurrent execution
- KU12.** the need for scheduler
- KU13.** context switching
- KU14.** the features of freeRTOS
- KU15.** the value proposition of freeRTOS
- KU16.** advance C covering memory management
- KU17.** different types of memory available
- KU18.** freeRTOS Heap Memory Management (dynamic vs static, predictability, fragmentation, different memory allocation schemes)
- KU19.** the freeRTOS Heap Utility Functions
- KU20.** the process of optimizing memory
- KU21.** the process of debugging memory
- KU22.** the memory map
- KU23.** the relevant concepts - architecture flat memory model, no distinction between kernel and user memory in RTOS, memory protection unit etc.
- KU24.** the concept of freeRTOS Tasks
- KU25.** the freeRTOS Tasks APIs
- KU26.** the process of creating tasks and setting task priorities
- KU27.** task state transitions
- KU28.** idle task and scheduling algorithms
- KU29.** time measurement and tick interrupt
- KU30.** Queue and freeRTOS Queue APIs
- KU31.** the use of data storage for Queue (data of fixed length, variable length, mem copy vs by reference etc.)
- KU32.** blocking read, and write in Queue
- KU33.** the process of receiving data from multiple queues
- KU34.** Events and Interrupt Service Routines (ISRs)
- KU35.** Tasks vs ISRs comparison
- KU36.** the process of Nested Interrupt handling
- KU37.** the concept of semaphores
- KU38.** the synchronization mechanisms of binary Semaphores
- KU39.** the use of counting semaphores
- KU40.** the problem of shared resource corruption (multi-tasking system)
- KU41.** the read, modify, write operations
- KU42.** the Atomic operations with respect to resource management

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- KU43.** the re-entrant functions or thread safe functions
- KU44.** the concept and conditions of Mutual Exclusion
- KU45.** Critical Section with respect to shared resources
- KU46.** the Mutex (locking) mechanism
- KU47.** the deadlocks and starvation in programming
- KU48.** Priority inversion
- KU49.** Priority Inheritance
- KU50.** the utility of gatekeeper task (for guarding shared resources)
- KU51.** the benefit of event groups for multiple task synchronization
- KU52.** Event groups for multiple task synchronization
- KU53.** the context and benefit of Software Timers
- KU54.** the process of creating software timer
- KU55.** the requirement and process of resetting of software timer
- KU56.** the use of hardware timers and watchdogs
- KU57.** tasks communication objects vs direct task notifications
- KU58.** the benefits of task notifications
- KU59.** the use of SMP port of freeRTOS
- KU60.** the benefits of memory protection unit support
- KU61.** the requirement of low power support (Ticket execution feature)
- KU62.** trace and runtime stats
- KU63.** the requirement of refresh on device drivers and integration
- KU64.** different popular RTOSes
- KU65.** the file system concept
- KU66.** Free RTOS , FAT (FAT12, FAT16, FAT32 DOS/Windows compatible File System)
- KU67.** File System APIs, source code organization, and configuration
- KU68.** different examples of FAT, e.g. basic file use, stdio API tests, creating a disk etc.
- KU69.** examples of FTP server and HTTP Web Server
- KU70.** system design from the perspective of hardware, tasks, interrupts/events, timers, and real-time processing
- KU71.** the common problems encountered (crash, ISR issues, stack overflow issues), debugging (asserts, traces/logs), and profiling, etc.
- KU72.** project delivery based on freeRTOS and embedded hardware board or emulator
- KU73.** how to configure and run simple programs on the emulator and on the board
- KU74.** target board and GPIO programming
- KU75.** the need of Interrupts
- KU76.** the need for Generic Clock tree, CPU interrupt flow, and Interrupt flow
- KU77.** Interrupt Flow inside ISR, and Hardware ISR Coding practice
- KU78.** the process of developing and testing a simple serial Driver
- KU79.** the need of Kernel drivers
- KU80.** the benefits of Loadable modules

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- KU81.** the process of writing a Hello World Device driver
- KU82.** the configuration of a USB device in Windows/Linux environment
- KU83.** what a network driver is
- KU84.** the configurability of Network drivers (e.g. how SSID is configured for Wi-Fi, or Ethernet is put in promiscuous mode)
- KU85.** what optimization is
- KU86.** different optimization methods
- KU87.** Elaborate on the configurability of Network drivers (e.g. how SSID is configured for Wi-Fi, or Ethernet is put in promiscuous mode)
- KU88.** different optimization methods

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** listen attentively to understand the information/ instructions being shared by the speaker
- GS3.** communicate clearly and politely with co-workers and clients
- GS4.** coordinate with co-workers to achieve work objectives
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify possible disruptions to work and take appropriate preventive measures
- GS7.** take quick decisions to deal with workplace emergencies/ accidents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Embedded OS</i>	3	9	-	-
PC1. Elaborate on freeRTOS	-	1	-	-
PC2. Explain the process of downloading and installing freeRTOS	-	1	-	-
PC3. Describe the use of Compilation and build tool setup	-	-	-	-
PC4. Explain the simulator/ emulator requirements	1	1	-	-
PC5. Elaborate on hardware setup and demo	-	1	-	-
PC6. Explain freeRTOS source code organization	1	1	-	-
PC7. Explain the use of freeRTOS ports	-	1	-	-
PC8. Describe the use of demo applications	-	1	-	-
PC9. Describe the process of configuring freeRTOS	1	-	-	-
PC10. Describe the process of building a simple demo application	-	1	-	-
PC11. Explain the applicable simulator/ emulator requirements	-	1	-	-
<i>Concept of Real-time OS</i>	1	4	-	-
PC12. Compare hard real time with soft real time	-	1	-	-
PC13. Elaborate on Multi-threading/ Multi-tasking / Concurrent execution	1	1	-	-
PC14. Explain the need for scheduler	-	-	-	-
PC15. Explain context switching	-	-	-	-
PC16. Identify the features of freeRTOS	-	1	-	-
PC17. Explain the value proposition of freeRTOS	-	1	-	-
<i>Memory Management</i>	3	7	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18. Elaborate on advance C covering memory management	1	1	-	-
PC19. Identify different types of memory available	-	1	-	-
PC20. Elaborate on freeRTOS Heap Memory Management (dynamic vs static, predictability, fragmentation, different memory allocation schemes)	-	1	-	-
PC21. Explain the freeRTOS Heap Utility Functions	-	1	-	-
PC22. Describe the process of optimizing memory	1	1	-	-
PC23. Describe the process of debugging memory	-	1	-	-
PC24. Elaborate on memory map	1	-	-	-
PC25. Explain the relevant concepts - architecture flat memory model, no distinction between kernel and user memory in RTOS, memory protection unit etc.	-	1	-	-
<i>Task Management</i>	2	5	-	-
PC26. Explain the concept of freeRTOS Tasks	-	1	-	-
PC27. Elaborate on freeRTOS Tasks APIs	-	1	-	-
PC28. Describe the process of creating tasks and setting task priorities	-	1	-	-
PC29. Elaborate on task state transitions	1	1	-	-
PC30. Elaborate on idle task and scheduling algorithms	-	1	-	-
PC31. Elaborate on time measurement and tick interrupt	1	-	-	-
<i>Inter Task (Process) Communication (synchronization)</i>	2	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC32. Elaborate on Queue and freeRTOS Queue APIs	-	1	-	-
PC33. Explain the use of data storage for Queue (data of fixed length, variable length, mem copy vs by reference etc.)	-	1	-	-
PC34. Elaborate on blocking read, and write in Queue	-	1	-	-
PC35. Describe the process of receiving data from multiple queues	1	1	-	-
PC36. Elaborate on Events and Interrupt Service Routines (ISRs)	-	1	-	-
PC37. Conduct Tasks vs ISRs comparison	-	1	-	-
PC38. Describe the process of Nested Interrupt handling	-	-	-	-
PC39. Explain the concept of semaphores	1	1	-	-
PC40. Explain the synchronization mechanisms of binary Semaphores	-	1	-	-
PC41. Explain the use of counting semaphores	-	-	-	-
<i>Resource Management</i>	4	13	-	-
PC42. Explain the problem of shared resource corruption (multi-tasking system)	-	1	-	-
PC43. Describe the read, modify, write operations	1	1	-	-
PC44. Elaborate on Atomic operations with respect to resource management	-	1	-	-
PC45. Explain re-entrant functions or thread safe functions	-	1	-	-
PC46. Explain the concept and conditions of Mutual Exclusion	-	1	-	-
PC47. Elaborate on Critical Section with respect to shared resources	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC48. Elaborate on Critical Section	-	-	-	-
PC49. Explain the Mutex (locking) mechanism	-	1	-	-
PC50. Elaborate on deadlocks and starvation in programming	-	1	-	-
PC51. Explain Priority inversion	1	1	-	-
PC52. Elaborate on Priority Inheritance	-	1	-	-
PC53. Explain the utility of gatekeeper task (for guarding shared resources)	-	1	-	-
PC54. Explain the benefit of event groups for multiple task synchronization	1	1	-	-
PC55. Elaborate on event groups for multiple task synchronization	-	1	-	-
<i>Timer Management</i>	1	3	-	-
PC56. Elaborate on the context and benefit of Software Timers	-	1	-	-
PC57. Describe the process of creating software timer	1	-	-	-
PC58. Elaborate on the requirement and process of resetting of software timer	-	1	-	-
PC59. Explain the use of hardware timers and watchdogs	-	1	-	-
<i>Task Notifications</i>	-	2	-	-
PC60. Conduct tasks communication objects vs direct task notifications comparison	-	1	-	-
PC61. Explain the benefits of task notifications	-	1	-	-
<i>Advance RTOS concepts</i>	2	6	-	-
PC62. Explain the use of SMP port of freeRTOS	1	1	-	-
PC63. Explain the benefits of memory protection unit support	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC64. Explain the requirement of low power support (Ticket execution feature)	-	1	-	-
PC65. Elaborate on trace and runtime stats	-	1	-	-
PC66. Explain the requirement of refresh on device drivers and integration	-	1	-	-
PC67. Conduct comparison of different popular RTOSes	1	1	-	-
<i>RTOS File System (freeRTOS+FAT)</i>	1	3	-	-
PC68. Explain the file system concept	-	-	-	-
PC69. Elaborate on FreeRTOS + FAT (FAT12, FAT16, FAT32 DOS/Windows compatible File System)	-	1	-	-
PC70. Elaborate on File System APIs, source code organization, and configuration	1	1	-	-
PC71. List different examples of FAT, e.g. basic file use, stdio API tests, creating a disk etc.	-	1	-	-
PC72. List examples of FTP server and HTTP Web Server	-	-	-	-
<i>RTOS Project</i>	1	3	-	-
PC73. Explain system design from the perspective of hardware, tasks, interrupts/events, timers, and real-time processing	1	1	-	-
PC74. Identify the common problems encountered (crash, ISR issues, stack overflow issues), debugging (asserts, traces/logs), and profiling, etc.	-	1	-	-
PC75. Describe project delivery based on freeRTOS and embedded hardware board or emulator	-	1	-	-
<i>Introduction to ARM Cortex-M and its Architecture</i>	1	2	-	-
PC76. Explain how to configure and run simple programs on the emulator and on the board	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC77. Elaborate on target board and GPIO programming	1	1	-	-
<i>Interrupts, Timers and counters</i>	1	3	-	-
PC78. Explain the need of Interrupts	-	1	-	-
PC79. Elaborate on the need for Generic Clock tree, CPU interrupt flow, and Interrupt flow	-	1	-	-
PC80. Elaborate on Interrupt Flow - inside ISR, and Hardware ISR's - Coding practice	1	1	-	-
<i>Driver development for a baremetal environmen</i>	-	1	-	-
PC81. Describe the process of developing and testing a simple serial Driver	-	1	-	-
<i>Linux driver development and testing</i>	2	6	-	-
PC82. Elaborate on the need of Kernel drivers	-	1	-	-
PC83. Explain the benefits of Loadable modules	1	1	-	-
PC84. Describe the process of writing a Hello World Device driver	-	1	-	-
PC85. Describe the configuration of a USB device in Windows/Linux environment	1	1	-	-
PC86. Explain what a network driver is	-	1	-	-
PC87. Elaborate on the configurability of Network drivers (e.g. how SSID is configured for Wi-Fi, or Ethernet is put in promiscuous mode)	-	1	-	-
<i>Optimizations</i>	1	-	-	-
PC88. Explain what optimization is	-	-	-	-
PC89. Describe different optimization methods	1	-	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4603
NOS Name	Explain Embedded OS Fundamentals, Driver, Applications, and Middleware
Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

SSC/N4604: Describe the process of Verification & Validation

Description

This unit covers the fundamentals of testing, testing throughout the SDLC, testing techniques, test infrastructure, monitoring and control.

Scope

The scope covers the following :

- Fundamentals of testing
- Testing Throughout the Software Development Life
- Static Techniques
- Test Techniques
- Test Infrastructure
- Test Monitoring & Control

Elements and Performance Criteria

Fundamentals of testing

To be competent, the user/individual on the job must be able to:

- PC1.** Explain what testing is
- PC2.** Explain why testing is necessary
- PC3.** Explain the seven testing principles
- PC4.** Describe the test process
- PC5.** Identify the psychological factors that influence the success of testing

Testing Throughout the Software Development Life

To be competent, the user/individual on the job must be able to:

- PC6.** Identify different approaches to testing required for different software development lifecycle models
- PC7.** Compare the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities
- PC8.** Compare different types of tests, e.g. functional, non-functional, and white-box testing

Static Techniques

To be competent, the user/individual on the job must be able to:

- PC9.** Describe the value of static testing using appropriate examples
- PC10.** Summarize the activities of the work product review process

Test Techniques

To be competent, the user/individual on the job must be able to:

- PC11.** Identify different categories of test techniques
- PC12.** Elaborate on black-box test techniques
- PC13.** Elaborate on white-box test techniques
- PC14.** Describe the experience based test techniques

Qualification Pack

PC15. Explain the use of test design techniques for embedded systems

Test Infrastructure

To be competent, the user/individual on the job must be able to:

PC16. Identify various requirements of embedded software test environments, e.g. simulation, prototyping, pre-production, and post-production

PC17. Identify the test tools required for embedded software testing

PC18. Describe the categorization of test tools

PC19. Explain the appropriate functions in test automation

Test Monitoring & Control

To be competent, the user/individual on the job must be able to:

PC20. Describe the process of test monitoring and control

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. what testing is

KU2. why testing is necessary

KU3. the seven testing principles

KU4. the test process

KU5. the psychological factors that influence the success of testing

KU6. different approaches to testing required for different software development lifecycle models

KU7. the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities

KU8. different types of tests, e.g. functional, non-functional, and white-box testing

KU9. the value of static testing using appropriate examples

KU10. the activities of the work product review process

KU11. the different categories of test techniques

KU12. the black-box test techniques

KU13. the white-box test techniques

KU14. the experience based test techniques

KU15. the use of test design techniques for embedded systems

KU16. various requirements of embedded software test environments, e.g. simulation, prototyping, pre-production, and post-production

KU17. the test tools required for embedded software testing

KU18. the categorization of test tools

KU19. the appropriate functions in test automation

KU20. the process of test monitoring and control

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** maintain work-related notes and records
- GS2.** listen attentively to understand the information/ instructions being shared by the speaker
- GS3.** communicate clearly and politely with co-workers and clients
- GS4.** coordinate with co-workers to achieve work objectives
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify possible disruptions to work and take appropriate preventive measures
- GS7.** take quick decisions to deal with workplace emergencies/ accidents
- GS8.** evaluate all possible solutions to a problem to select the best one

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamentals of testing</i>	5	19	-	-
PC1. Explain what testing is	1	4	-	-
PC2. Explain why testing is necessary	1	3	-	-
PC3. Explain the seven testing principles	1	4	-	-
PC4. Describe the test process	1	4	-	-
PC5. Identify the psychological factors that influence the success of testing	1	4	-	-
<i>Testing Throughout the Software Development Life</i>	6	11	-	-
PC6. Identify different approaches to testing required for different software development lifecycle models	2	3	-	-
PC7. Compare the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities	2	4	-	-
PC8. Compare different types of tests, e.g. functional, non-functional, and white-box testing	2	4	-	-
<i>Static Techniques</i>	2	7	-	-
PC9. Describe the value of static testing using appropriate examples	1	3	-	-
PC10. Summarize the activities of the work product review process	1	4	-	-
<i>Test Techniques</i>	5	19	-	-
PC11. Identify different categories of test techniques	1	4	-	-
PC12. Elaborate on black-box test techniques	1	4	-	-
PC13. Elaborate on white-box test techniques	1	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Describe the experience based test techniques	1	3	-	-
PC15. Explain the use of test design techniques for embedded systems	1	4	-	-
<i>Test Infrastructure</i>	6	15	-	-
PC16. Identify various requirements of embedded software test environments, e.g. simulation, prototyping, pre-production, and post-production	2	4	-	-
PC17. Identify the test tools required for embedded software testing	1	3	-	-
PC18. Describe the categorization of test tools	1	4	-	-
PC19. Explain the appropriate functions in test automation	2	4	-	-
<i>Test Monitoring & Control</i>	1	4	-	-
PC20. Describe the process of test monitoring and control	1	4	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4604
NOS Name	Describe the process of Verification & Validation
Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

SSC/N4605: Explain Packaging, Release, and DevOps

Description

This unit covers various aspects of packaging, release, and DevOps, such as version control, Jenkins, continuous integration and delivery with Jenkins, etc.

Scope

The scope covers the following :

- DevOps Fundamentals
- Version Control with Git
- Packaging, Release and Continuous Integration
- Introduction to Jenkins
- Introduction to Jenkins

Elements and Performance Criteria

DevOps Fundamentals

To be competent, the user/individual on the job must be able to:

- PC1.** Describe different DevOps Stages, e.g. building, integration, deployment, monitoring, improvement
- PC2.** Explain the basics of virtualization
- PC3.** Explain the basic Linux Commands

Version Control with Git

To be competent, the user/individual on the job must be able to:

- PC4.** Explain the use of version control system
- PC5.** Explain the importance and use of Git and GitHub

Packaging, Release and Continuous Integration

To be competent, the user/individual on the job must be able to:

- PC6.** Explain the importance and benefits of building and packaging
- PC7.** Discuss the benefits of GNU Make and Makefiles in the building and packaging process
- PC8.** Explain the benefits of using Keil Microcontroller Development Kit (MDK) for developing microcontrollers

Introduction to Jenkins

To be competent, the user/individual on the job must be able to:

- PC9.** Explain the importance and benefits of Jenkins
- PC10.** Describe the procedure of setting up Jenkins
- PC11.** Describe the process of integrating Jenkins with GitHub

Continuous Integration and Delivery with Jenkins

To be competent, the user/individual on the job must be able to:

- PC12.** Describe the process of continuous integration and delivery with Jenkins

Qualification Pack

PC13. Explain the importance of build (GNU Make and makefiles) in Continuous Integration (Compile, Unit Tests, Build Package)

PC14. Describe the process of running Automated Tests (with simulation) and Reporting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different DevOps Stages, e.g. building, integration, deployment, monitoring, improvement

KU2. the basics of virtualization

KU3. the use of version control system

KU4. the importance and use of Git and GitHub

KU5. the importance and benefits of building and packaging

KU6. the importance and benefits of Jenkins

KU7. the process of continuous integration and delivery with Jenkins

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work-related notes and records

GS2. listen attentively to understand the information/ instructions being shared by the speaker

GS3. communicate clearly and politely with co-workers and clients

GS4. coordinate with co-workers to achieve work objectives

GS5. plan and prioritize tasks to ensure timely completion

GS6. identify possible disruptions to work and take appropriate preventive measures

GS7. take quick decisions to deal with workplace emergencies/ accidents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>DevOps Fundamentals</i>	5	16	-	-
PC1. Describe different DevOps Stages, e.g. building, integration, deployment, monitoring, improvement	2	6	-	-
PC2. Explain the basics of virtualization	1	5	-	-
PC3. Explain the basic Linux Commands	2	5	-	-
<i>Version Control with Git</i>	4	10	-	-
PC4. Explain the use of version control system	2	5	-	-
PC5. Explain the importance and use of Git and GitHub	2	5	-	-
<i>Packaging, Release and Continuous Integration</i>	5	17	-	-
PC6. Explain the importance and benefits of building and packaging	2	6	-	-
PC7. Discuss the benefits of GNU Make and Makefiles in the building and packaging process	2	6	-	-
PC8. Explain the benefits of using Keil Microcontroller Development Kit (MDK) for developing microcontrollers	1	5	-	-
<i>Introduction to Jenkins</i>	6	16	-	-
PC9. Explain the importance and benefits of Jenkins	2	5	-	-
PC10. Describe the procedure of setting up Jenkins	2	5	-	-
PC11. Describe the process of integrating Jenkins with GitHub	2	6	-	-
<i>Continuous Integration and Delivery with Jenkins</i>	5	16	-	-
PC12. Describe the process of continuous integration and delivery with Jenkins	2	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Explain the importance of build (GNU Make and makefiles) in Continuous Integration (Compile, Unit Tests, Build Package)	2	6	-	-
PC14. Describe the process of running Automated Tests (with simulation) and Reporting	1	5	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4605
NOS Name	Explain Packaging, Release, and DevOps
Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

SSC/N4606: Elaborate on embedded machine learning and Domain Applications

Description

This unit covers various aspects of machine learning and domain applications, e.g. Neural Networks and ML Model, Automotive Embedded Systems, Autosar Architecture and Model, etc.

Scope

The scope covers the following :

- Embedded machine learning
- Machine Learning & Embedded Systems
- Neural Networks and ML Model
- Embedded Machine Learning Practical
- Automotive Embedded Systems
- Autosar Architecture : Introduction
- AUTOSAR Model Deployment in MATLAB
- Project on Embedded Machine Learning and Automotive Embedded System

Elements and Performance Criteria

Embedded machine learning

To be competent, the user/individual on the job must be able to:

PC1. Elaborate the concept of machine learning

Machine Learning & Embedded Systems

To be competent, the user/individual on the job must be able to:

PC2. Explain the interconnection between machine learning on embedded systems

PC3. Elaborate on machine learning on microcontrollers

PC4. Describe the process of dataset and anomaly detection

PC5. Discuss the criteria for feature selection and extraction

PC6. Explain how to deploy a trained model to RPI/Arduino

Neural Networks and ML Model

To be competent, the user/individual on the job must be able to:

PC7. Elaborate on neural networks and training methods Model Evaluation Techniques Overfitting and Underfitting

PC8. Describe the model evaluation techniques

PC9. Explain the difference between overfitting and underfitting

PC10. Elaborate on how to deploy a model to an embedded system (RPI Board)

Embedded Machine Learning Practical

To be competent, the user/individual on the job must be able to:

PC11. Describe the process of feature extraction for audio data convolutional neural networks

PC12. Elaborate on audio classification and sampling audio signals MFCCs and CNNs

Qualification Pack

PC13. Elaborate on demo application on audio class inferencing

Automotive Embedded Systems

To be competent, the user/individual on the job must be able to:

PC14. Discuss the importance of automotive embedded systems

PC15. identify different domains in automotive systems

PC16. Describe the process of model based Development in MATLAB

PC17. Elaborate on software component modelling simulation & code generation

Autosar Architecture : Introduction

To be competent, the user/individual on the job must be able to:

PC18. Define AUTOSAR architecture

PC19. Elaborate on MCAL layer and services layer

PC20. Elaborate on autosar diagnostics & memstack

AUTOSAR Model Deployment in MATLAB

To be competent, the user/individual on the job must be able to:

PC21. Explain the importance of Modelling Autosar SWCs in MATLAB

PC22. Identify the difference between Embedded Coder and Autosar Coder

PC23. Elaborate on Autosar Editor – Code Mapping

PC24. Elaborate on Functional Safety in an AUTOSAR model

Project on Embedded Machine Learning and Automotive Embedded System

To be competent, the user/individual on the job must be able to:

PC25. Elaborate on real-time classification of audio streams

PC26. Explain modelling in automotive embedded systems

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the concept of machine learning

KU2. the interconnection between machine learning on embedded systems

KU3. machine learning on microcontrollers

KU4. the process of dataset and anomaly detection

KU5. the criteria for feature selection and extraction

KU6. how to deploy a trained model to RPI/Arduino

KU7. neural networks and training methods Model Evaluation Techniques Overfitting and Underfitting

KU8. the model evaluation techniques

KU9. the difference between overfitting and underfitting

KU10. how to deploy a model to an embedded system (RPI Board)

KU11. the process of feature extraction for audio data convolutional neural networks

KU12. audio classification and sampling audio signals MFCCs and CNNs

KU13. demo application on audio class inferencing

Qualification Pack

- KU14.** the importance of automotive embedded systems
- KU15.** different domains in automotive systems
- KU16.** the process of model based Development in MATLAB
- KU17.** software component modelling simulation & code generation
- KU18.** AUTOSAR architecture
- KU19.** MCAL layer and services layer
- KU20.** autosar diagnostics & memstack
- KU21.** the importance of Modelling Autosar SWCs in MATLAB
- KU22.** the difference between Embedded Coder and Autosar Coder
- KU23.** Autosar Editor Code Mapping
- KU24.** Functional Safety in an AUTOSAR model
- KU25.** the real-time classification of audio streams
- KU26.** modelling in automotive embedded systems

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** listen attentively to understand the information/ instructions being shared by the speaker
- GS3.** communicate clearly and politely with co-workers and clients
- GS4.** coordinate with co-workers to achieve work objectives
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify possible disruptions to work and take appropriate preventive measures
- GS7.** take quick decisions to deal with workplace emergencies/ accidents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Embedded machine learning</i>	1	3	-	-
PC1. Elaborate the concept of machine learning	1	3	-	-
<i>Machine Learning & Embedded Systems</i>	5	15	-	-
PC2. Explain the interconnection between machine learning on embedded systems	1	3	-	-
PC3. Elaborate on machine learning on microcontrollers	1	3	-	-
PC4. Describe the process of dataset and anomaly detection	1	3	-	-
PC5. Discuss the criteria for feature selection and extraction	1	3	-	-
PC6. Explain how to deploy a trained model to RPI/Arduino	1	3	-	-
<i>Neural Networks and ML Model</i>	3	11	-	-
PC7. Elaborate on neural networks and training methods Model Evaluation Techniques Overfitting and Underfitting	1	3	-	-
PC8. Describe the model evaluation techniques	-	3	-	-
PC9. Explain the difference between overfitting and underfitting	1	2	-	-
PC10. Elaborate on how to deploy a model to an embedded system (RPI Board)	1	3	-	-
<i>Embedded Machine Learning Practical</i>	3	9	-	-
PC11. Describe the process of feature extraction for audio data convolutional neural networks	1	3	-	-
PC12. Elaborate on audio classification and sampling audio signals MFCCs and CNNs	1	3	-	-
PC13. Elaborate on demo application on audio class inferencing	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Automotive Embedded Systems</i>	4	11	-	-
PC14. Discuss the importance of automotive embedded systems	1	3	-	-
PC15. identify different domains in automotive systems	1	3	-	-
PC16. Describe the process of model based Development in MATLAB	1	2	-	-
PC17. Elaborate on software component modelling simulation & code generation	1	3	-	-
<i>Autosar Architecture : Introduction</i>	3	9	-	-
PC18. Define AUTOSAR architecture	1	3	-	-
PC19. Elaborate on MCAL layer and services layer	1	3	-	-
PC20. Elaborate on autosar diagnostics & memstack	1	3	-	-
<i>AUTOSAR Model Deployment in MATLAB</i>	4	11	-	-
PC21. Explain the importance of Modelling Autosar SWCs in MATLAB	1	3	-	-
PC22. Identify the difference between Embedded Coder and Autosar Coder	1	2	-	-
PC23. Elaborate on Autosar Editor – Code Mapping	1	3	-	-
PC24. Elaborate on Functional Safety in an AUTOSAR model	1	3	-	-
<i>Project on Embedded Machine Learning and Automotive Embedded System</i>	2	6	-	-
PC25. Elaborate on real-time classification of audio streams	1	3	-	-
PC26. Explain modelling in automotive embedded systems	1	3	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N4606
NOS Name	Elaborate on embedded machine learning and Domain Applications
Sector	IT-ITeS
Sub-Sector	Engineering R&D
Occupation	Software Development
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

SSC/N9014: Maintain an inclusive, environmentally sustainable workplace

Description

The unit is about implementing and improving diversity equality and inclusion in a sustainable and environment friendly workplace.

Scope

The scope covers the following :

- Sustainable Practices
- Respect diversity and strengthen practices to promote equity (equality)/inclusivity

Elements and Performance Criteria

Sustainable Practices

To be competent, the user/individual on the job must be able to:

- PC1.** optimize usage of electricity/energy, materials, and water in various asks / activities / processes and plan the implementation of energy efficient systems in a phased manner
- PC2.** segregate recyclable, non-recyclable and hazardous waste generated for disposal or efficient waste management

Respect diversity and strengthen practices to promote equity (equality)/inclusivity

To be competent, the user/individual on the job must be able to:

- PC3.** understand the diversity policy of the organization and use internal & external communication to colleagues to improve
- PC4.** comply with PwD inclusive policies for an adaptable and equitable work environment
- PC5.** improve through specifically designed recruitment practices, PwD friendly infrastructure, job roles, etc.
- PC6.** use and advocate for appropriate verbal/nonverbal communication, schemes and benefits of PwD.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization's policies and procedures about gender inclusivity, equality and sustainability while working with colleagues and your role and responsibilities in relation to this
- KU2.** inclusive tools and practices of communication to acknowledge/validate, share and promote the cause of gender parity at workplace. For example - supporting women with mentorship programs, speaking out against discriminatory practices or harassment
- KU3.** the concept of gender, gender equality and gender discrimination, and all forms of gender discrimination, violence and inequality, including the current and historical causes of gender inequality in the workplace

Qualification Pack

- KU4.** how to maintain and provide a conducive work environment that is free from any harassment. facilities and amenities to PwD to perform and excel in their role
- KU5.** organization's redressal mechanisms (like the POSH committee) to address harassment and bias at the workplace, with awareness of prevalent legislations against bias and sexual harassment
- KU6.** initiatives towards efficient use of natural resources and energy, reduction and prevention of pollution and promoting waste avoidance and recycling measures in line with internationally disseminated technologies and practices
- KU7.** all about various energy options including renewable and non-renewable with their environmental impacts, health issues, usage, safety and energy security
- KU8.** implications that any non-compliance with electricity and energy may have on individuals and the organization
- KU9.** the organization's electricity first aid emergency procedures
- KU10.** how to monitor, measure and report performance of environmental conservation
- KU11.** different types of electricity accidents, safety and security and how and when to report these
- KU12.** how to use the electricity/energy safety, accident reporting, emergency procedures and the importance of these

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read PwD instructions, guidelines, procedures, diversity policies/acts, rules and service level agreements
- GS2.** be aware of one's own gender identity and gender role and respectful of the gender performances of others
- GS3.** organize team building or sensitization workshops to address gender biases, stereotypes and potentially blind spots
- GS4.** clarify personal norms and values related to energy production and usage as well as to reflect and evaluate their own energy usage in terms of efficiency and sufficiency
- GS5.** listen and communicate (oral) effectively and accurately on all PwD policies
- GS6.** apply balanced judgments in gender diversity situations
- GS7.** take action to reduce the carbon footprint of business activities and embed environmental responsibility
- GS8.** calibration session with employees to discuss gender biases, stereotypes and potentially blind spots

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sustainable Practices</i>	10	30	-	-
PC1. optimize usage of electricity/energy, materials, and water in various asks / activities / processes and plan the implementation of energy efficient systems in a phased manner	5	15	-	-
PC2. segregate recyclable, non-recyclable and hazardous waste generated for disposal or efficient waste management	5	15	-	-
<i>Respect diversity and strengthen practices to promote equity (equality)/inclusivity</i>	10	50	-	-
PC3. understand the diversity policy of the organization and use internal & external communication to colleagues to improve	5	10	-	-
PC4. comply with PwD inclusive policies for an adaptable and equitable work environment	-	10	-	-
PC5. improve through specifically designed recruitment practices, PwD friendly infrastructure, job roles, etc.	-	20	-	-
PC6. use and advocate for appropriate verbal/nonverbal communication, schemes and benefits of PwD.	5	10	-	-
NOS Total	20	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N9014
NOS Name	Maintain an inclusive, environmentally sustainable workplace
Sector	IT-ITeS
Sub-Sector	IT Services, Business Process Management, Engineering R&D, Software Product Development, Future Skills
Occupation	Generic,
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	22/01/2022
Next Review Date	22/01/2025
NSQC Clearance Date	22/01/2022

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification File will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective / option NOS/set of NOS
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below)
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion
6. To pass a QP, a trainee should score an average of 70% or more
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification File.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N4601.Explain the Software Development Lifecycle (SDLC) and agile methodology	25	75	0	0	100	14
SSC/N4602.Elaborate on Hardware, Lab Familiarization and Advanced C	25	75	0	0	100	14
SSC/N4603.Explain Embedded OS Fundamentals, Driver, Applications, and Middleware	25	75	0	0	100	14
SSC/N4604.Describe the process of Verification & Validation	25	75	0	0	100	14
SSC/N4605.Explain Packaging, Release, and DevOps	25	75	0	0	100	14
SSC/N4606.Elaborate on embedded machine learning and Domain Applications	25	75	0	0	100	14
SSC/N9014.Maintain an inclusive, environmentally sustainable workplace	20	80	-	-	100	6
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	190	560	-	-	750	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Helpdesk	Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.